

Deep Avance mini-project

Assignment Report

Eng. Robert Bejan

Eng. Teodor Butnaru

I. Introduction

This paper explores how to translate images from one class to another using the Animal Faces HQ (AFHQ) dataset. We compared three popular types of AI models: Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models. Our experiments show clear differences between them. VAEs often produced blurry images, while GANs, specifically CycleGAN, did a good job of changing animal traits, though they still had limitations. Diffusion Models were the clear winner, creating the sharpest and most realistic images. However, they are not perfect yet. The following sections describe how we built these models, compare the results, and suggest ways to fix these issues in the future.

II. Dataset Animal Faces HQ (AFHQ)

The AFHQ dataset consists of 15,000 high-quality images at a consistent resolution of 512×512 pixels. It is divided into three distinct domains: Cat, Dog, and Wild, with each category containing 5,000 images. Unlike many other datasets, AFHQ offers a high degree of intra-class diversity; each domain includes more than eight different breeds or species, ensuring a wide range of features, textures, and appearances. In the current task of image-to-image translation, we have chosen only the two specifically defined classes in the database, Cat and Dog.

III. Approaches

III.a. Methodology and Model Selection

The selection of generative architectures for this project was fundamentally guided by the theory explored during our coursework. We identified three primary paradigms, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Diffusion Models, as the most robust candidates for addressing the complexities of the Animal Faces dataset. Each of these models offers a unique approach to the image-to-image translation problem: VAEs provide a structured latent space for feature manipulation, GANs utilize adversarial training for high-level detail preservation, and Diffusion Models offer state-of-the-art iterative refinement. By benchmarking these three distinct methodologies, we aimed to evaluate the trade-offs between reconstruction fidelity, training stability, and stylistic accuracy.

III.b. Conditional Variational Autoencoders (cVAE)

Our investigation into likelihood-based models focused on the Conditional VAE (cVAE), which extends the standard VAE by conditioning the latent space on domain labels to enable controlled class-to-class transformations. To optimize performance and analyze the impact of feature extraction, we developed two distinct architectural variations. The first was a baseline cVAE trained from scratch, where the encoder and decoder were initialized randomly, forcing the model to learn hierarchical animal features solely from the AFHQ dataset without prior knowledge.

For the second variation, we replaced the standard encoder with a pretrained SqueezeNet backbone. This allowed us to test whether leveraging the efficient, pre-learned feature representations of a lightweight classification network could yield a more structured latent space.

By comparing these approaches, we aimed to determine if transfer learning could improve the density and disentanglement of the latent representation, thereby generating sharper reconstructions than the baseline model.

III.c. Adversarial Learning and the Cycle-Consistency Constraint

Beyond the likelihood-based approach of VAEs, we explored Generative Adversarial Networks (GANs) to achieve better translations on the animal faces. We specifically implemented CycleGAN, as it is uniquely suited for the unpaired nature of the AFHQ dataset, where no direct one-to-one mapping exists between individual animals across classes. The architecture consists of two sets of generators and discriminators that engage in a min-max game to learn the underlying distribution of each domain.

To prevent the model from simply hallucinating unrelated images, we utilized the cycle-consistency loss, a specific part of CycleGAN, which enforces the constraint that an image translated from the source to the target domain (e.g., cat to dog) must be reconstructible back to its original form when passed through the inverse generator. The bidirectional mapping should ensure that the model preserves the structural identity of the animal, such as head pose and eye position, while successfully morphing domain-specific features like snout shape, fur texture, and ear morphology. In our experiments, this adversarial framework significantly performed better than VAE-based models in generating satisfactory translations but with visible limitations given by the architecture.

III.d. Diffusion Model

To address the training instability and mode collapse often associated with GANs, we implemented a Conditional Denoising Diffusion Probabilistic Model (DDPM). Unlike adversarial methods, this framework learns to iteratively reverse a gradual noise-adding process, offering a more stable training objective.

Our architecture is built on a Conditional U-Net, but we introduced two critical enhancements to improve generation quality:

1. **Self-Attention Mechanisms:** We integrated self-attention layers at the lower resolutions of the U-Net. This allowed the model to process the image globally rather than just focusing on local pixels. In practice, this significantly improved geometric coherence; for example, it ensured that generated features like eyes were properly aligned and symmetric, preventing the disjointed look that earlier iterations suffered from.
2. **Exponential Moving Average (EMA):** We implemented EMA during training, which maintains a smoothed version of the model parameters. This greatly stabilized the training dynamics and, in our inference tests, produced images with fewer artifacts and better overall structure compared to using the raw model weights.

For conditioning, we moved beyond simple class labels by using rich, 512-dimensional text embeddings from a frozen CLIP encoder. These embeddings inject semantic concepts (like "a dog") into the network via Feature-wise Linear Modulation (FiLM) layers.

Finally, for the translation task itself, we utilized the SDEdit technique with Classifier-Free Guidance. By partially corrupting the source image with noise, we erase fine-grained details while keeping the overall shape. The model then reconstructs the image using the target class embedding. This semantic guidance, combined with the stability of EMA and the coherence from self-attention, yielded the sharpest and most consistent results in our study.

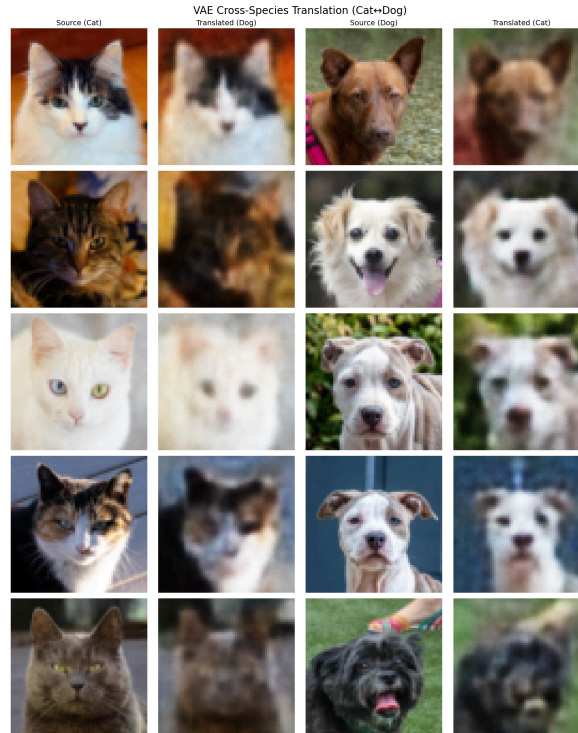
IV. Results

IV.a. cVAE trained from scratch

Training Parameters:

- **Training Duration:** We trained for 40 epochs using the **AdamW** optimizer with a weight decay of 10^{-4} .
- **Latent Space:** The model utilized a 256-dimensional latent space, providing a compressed representation for animal features.
- **Loss Function:** The objective function combined **Mean Squared Error (MSE)** for with a **KLD** loss for latent space regularization.

Visualization of the final image translation:

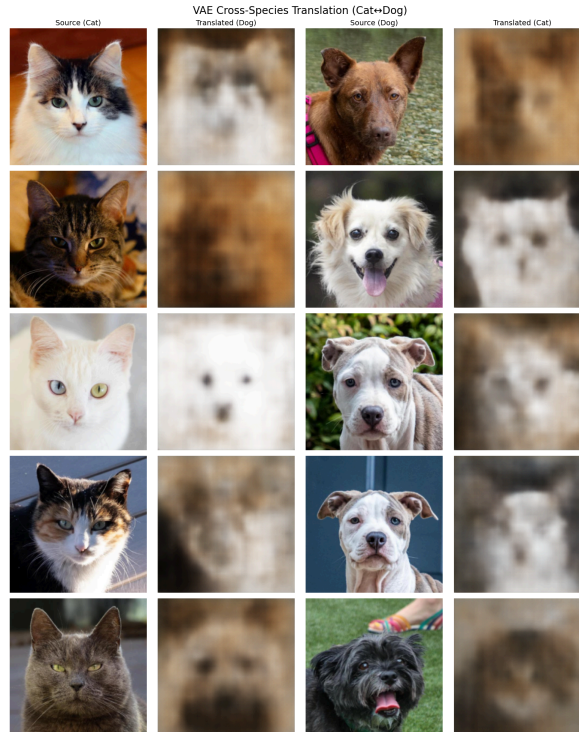


IV.a. cVAE trained with pretrained SqueezeNet encoder

Training Parameters:

- **Training Duration:** We trained for 50 epochs using the **AdamW** optimizer with a weight decay of 10^{-4} .
- **Latent Space:** The model utilized a 256-dimensional latent space, providing a compressed representation for animal features.
- **Loss Function:** The objective function combined **Mean Squared Error (MSE)** for with a **KLD** loss for latent space regularization.

Visualization of the final image translation:



IV.b. CycleGAN

Training Parameters:

- **Training Duration and Cycle-Consistency weight:** The models were trained for 200 epochs, using a cycle-consistency weight of 10 to ensure that the translated images retained the underlying geometry of the source animal.
- **Objective Function:** The total generator loss was composed of the adversarial GAN loss sum with the cycle-consistency loss.
- **Discriminator Stabilization:** To prevent the discriminators from overpowering the generators, we applied a 0.5 scaling factor to their respective loss functions.
- **Experience Replay:** An ImagePool of size 50 was implemented for both domains. This buffer stores previously generated images and provides them to the discriminators during training, reducing model oscillation and improving adversarial stability.

- **Optimizer:** We used the **Adam** optimizer with independent updates for the generator and discriminator networks.

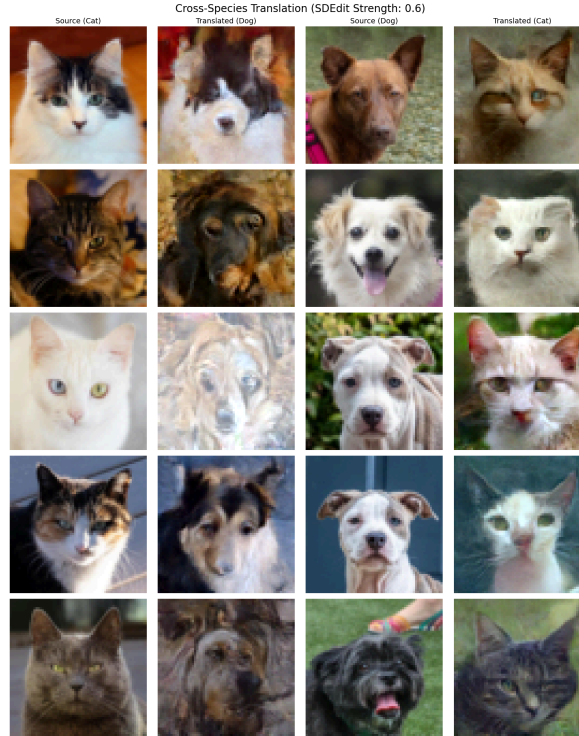
Visualization of the final image translation:



IV.c. Diffusion Model

Training Parameters:

- **Training Duration:** We trained an initial crude architecture for 700 epochs and after reinitialized with these weights for every architecture change and trained for 200 epochs using the **AdamW optimizer** with a learning rate of 2×10^{-4} . To stabilize the training dynamics, we utilized **Exponential Moving Average (EMA)** on the model weights with a decay of 0.9999.
- **Conditioning & Latents:** Instead of a VAE-style latent bottleneck, the model is conditioned on **512-dimensional CLIP text embeddings**. These embeddings are projected and injected into the network via **FiLM (Feature-wise Linear Modulation) layers**, allowing semantic concepts (like "Dog" or "Cat") to guide the generation at every resolution.
- **Noise Schedule:** The diffusion process operates over **1000 timesteps** using a **Cosine Beta Schedule**, which we found preserved information better than a linear schedule during the noise-addition phase.
- **Loss Function:** The objective function was a simple **Mean Squared Error (MSE)** calculated between the *predicted noise* and the *actual Gaussian noise* added to the image.



V. Conclusion

In this report, we investigated three different approaches to image-to-image translation on the AFHQ dataset: CVAEs, CycleGANs, and Diffusion Models. Our goal was to successfully translate features between animal classes (e.g., cat to dog) while maintaining the original image's structure.

Our experiments with **Variational Autoencoders (CVAE)** highlighted the difficulty of this task. The CVAE trained from scratch failed to perform the translation, merely producing a blurry reconstruction of the original input. Adding a pretrained EfficientNet encoder helped the model successfully translate species features, turning cats into distinguishable dogs, but the results remained very blurry. We acknowledge that this comparison was not entirely fair, as the pretrained model attempted to generate much larger images (224×224) compared to the 64×64 resolution used for other models, which significantly increased the difficulty.

CycleGAN demonstrated a slight improvement over the VAEs. It successfully maintained the subject's posture while transforming the species, producing much sharper images. However, it had noticeable limitations: distinct features like specific color patterns sometimes disappeared, and facial features were occasionally mushed or distorted during the morphing process.

Finally, the **Diffusion Model** achieved the best results in our study. It excelled at preserving high-level details that other models lost, such as color patterns, fur texture, and background context. While it was the clear winner, it was not perfect. We observed artifacts where eyes were missing or features became indistinguishable. This was especially common with dogs, likely because dog breeds have much higher visual variety compared to the relative uniformity of cats. Additionally, the low resolution (64×64) likely made it harder for the model to generate small details like eyes accurately.

Overall, while Diffusion Models provided the highest quality translation, future work is needed to handle diverse animal breeds better and scale up to higher resolutions to fix the hallucination of facial features.